Profile B — Open-Source / Kubernetes-Centric Sovereign

Focus: Maximum portability · CNCF-first · Zero vendor lock-in · Self-managed operational discipline

CNCF · v1.0

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COMPUTE / PLATFORM LAYER

portable across any IaaS / on-prem

RKE2 / k3s	Knative	ArgoCD	Istio	Temporal	Next.js (self-hosted)
CNCF-certified K8s	Serverless on K8s	GitOps CD	Service mesh	Durable workflows	SSR frontend
self-managed · any cloud	scale-to-zero	declarative · auditable	mTLS · traffic shifting	audit-replay native	containerized
30% lower compute than managed	50% cost saving for spiky webhooks	deploy time < 90s P95	canary p95 regression < 2%	workflow loss = 0 (durable log)	cloud-agnostic build artifacts

INTEGRATION / DATA LAYER

self-managed on K8s · open standards

Apache Kafka (Strimzi)	PostgreSQL (Crunchy PGO)	ClickHouse (Altinity)	Redis (Bitnami HA)	MinIO	Debezium CDC
K8s-operator-managed Kafka	Operator-managed Postgres	Self-managed OLAP	Sentinel + Cluster	S3-compatible object store	CDC · ingestion
30% lower cost at >1M msg/s	Patroni HA · PITR	sub-second on 100M rows	sub-ms P99 cache hit	runs on-prem OR cloud	Kafka Connect
portable across clouds	40% cheaper at scale vs Aurora	ReplicatedMergeTree · K8s	no vendor-specific ops	identical API to S3	schema evolution safe

SECURITY LAYER

open standards · auditable · portable

Keycloak	HashiCorp Vault	OPA (Open Policy Agent)	SPIFFE / SPIRE	Trivy + Falco	cert-manager + Let's Encrypt
Identity broker	Secrets · PKI	Policy-as-code	Workload identity	Scanning · runtime	TLS automation
OIDC · SAML · SCIM	dynamic secrets	sidecar enforcement	mTLS certificates	CVE + intrusion detect	ACME v2
no per-MAU license	audit log built-in	CMMI audit evidence	zero-trust east-west	CI gate on every PR	zero-touch cert rotation

OBSERVABILITY LAYER

OTel-native · self-hosted backends

Prometheus + Thanos	Grafana + Loki + Tempo	OpenTelemetry Collector	Alertmanager	Kubecost + OpenCost
Metrics · long-term store	Dashboards · logs · traces	Vendor-neutral SDK	Alerting	FinOps for K8s
cost per metric 60% lower	unified observability	zero vendor lock-in	SLO burn-rate alerts	per-tenant cost allocation
vs managed APM	single pane of glass	portable to A or C	p95 ack < 3 min	CMMI continuous improvement

CMMI LEVEL 5 TRADE-OFF (Profile B)

Selected for: **portability + zero lock-in** . Every component is CNCF or open-source; runs identically on any IaaS or on-prem. Sacrifices: 2–3x more platform engineering headcount; operational risk if the team is thin.

Quantitative basis: 30% lower compute cost than managed equivalents at >1M msg/s; 40% lower DB cost at scale; but ~18-month higher "time to first production change" without strong SRE bench.

Reversibility condition: if SRE headcount < 6 OR MTTD exceeds 30 min for two consecutive quarters‼ fall back to Profile A managed services for the failing component only (never the whole stack).